AI Failures in Legal Research and Graph-Based Verification: A Detailed Review of Key Papers

Prepared as a research literature report for the project *Guarded Browsing Agents for High-Stakes Legal Research*. The report summarises the data, methods, findings, and research-level implications of major studies on legal hallucination, citation verification, retrieval-augmented generation, and legal-agent evaluation.

1. Purpose and scope

This report reviews six research papers that are directly relevant to the problem of AI systems producing plausible but legally unreliable outputs: fabricated authorities, incorrect citations, misapplied rules, stale law, unsupported conclusions, and failures caused by restricted access to authoritative sources. The papers were selected because they collectively cover the full pipeline: standalone LLM behaviour, retrieval-augmented legal research tools, citation-graph verification, claim-level grounding, and multi-step browsing agents.

The report is organised paper by paper. For each paper, it explains the research question, dataset construction, experimental setup, principal results, error analysis, limitations, and the observations most useful for designing a legal knowledge graph and graph neural network (GNN). The final section synthesises these findings into a concrete architecture for training and evaluating a GNN over a versioned legal graph.

2. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models

Authors: Matthew Dahl, Varun Magesh, Mirac Suzgun, and Daniel E. Ho.
Publication: Journal of Legal Analysis, 2024; arXiv:2401.01301.
Core question: How frequently do general-purpose LLMs hallucinate when answering verifiable questions about U.S. federal case law, and which legal tasks are most vulnerable?

2.1 Research problem

The paper examines whether large language models can reliably answer factual questions about real federal court cases. Its central concern is not general factual hallucination but legal hallucination: an output that appears fluent and legally plausible while misstating a case, citation, holding, procedural fact, or doctrinal relationship.

The authors distinguish between errors internal to a model's reasoning and errors relative to external legal reality. This distinction matters because a model may generate an internally coherent legal explanation while citing a nonexistent case, attributing a holding to the wrong court, or describing a case in a way that the actual opinion does not support.

2.2 Data used

The study uses a large, systematically sampled corpus of U.S. federal judicial decisions. The authors sample 5,000 cases from each of three federal-court levels: the U.S. Supreme Court, the U.S. Courts of Appeals, and the U.S. District Courts. This produces a broad sample across 15,000 cases.

The paper constructs 14 evaluation tasks across three levels of complexity:

- **Low-complexity tasks:** questions about basic case metadata, such as case name, citation, court, and year.
- **Moderate-complexity tasks:** questions requiring extraction or recognition of information contained in the case, such as the disposition, parties, or factual setting.
- **High-complexity tasks:** questions requiring doctrinal reasoning, including whether one case agrees or disagrees with another, the year a case was overruled, the procedural posture, and the holding.

Ground truth is constructed using established legal-reference resources, including the Supreme Court Database, the Appeals Courts Database, the Library of Congress, and Shepard's citation services. The dataset and replication materials were released through the Harvard Dataverse and Hugging Face.

2.3 Method

The authors evaluate several widely used LLMs, including GPT-3.5, GPT-4, PaLM 2, and Llama 2. Each model is asked verifiable questions about sampled federal cases. Responses are then compared against external legal-reference sources rather than judged solely by another language model.

The evaluation is designed to test both retrieval-like factual recall and higher-order legal reasoning. For example, a model may be asked to identify the disposition of a case, determine whether two cases agree or conflict, identify the year a precedent was overruled, or state the holding of a case.

The paper also examines whether models can recognise their own uncertainty. It tests whether a model's expressed confidence correlates with actual correctness, and whether models can detect false premises embedded in a prompt.

2.4 Principal findings

- Legal hallucinations are pervasive. The study reports hallucination rates of at least 58% across the tested models, with rates reaching 88% for Llama 2 on certain verifiable federal-case questions.
- Performance deteriorates as legal complexity increases. Models perform relatively better on simple metadata questions but become substantially less reliable on questions involving quotations, dispositions, cited authorities, doctrinal agreement, procedural posture, subsequent history, and holdings.

- Models often fail to correct false premises. When a prompt contains an incorrect legal premise, models frequently produce an answer consistent with the false premise rather than identifying and correcting the error.
- Models cannot reliably identify their own hallucinations. Confidence is weakly related, at best, to correctness; a model may express high confidence while producing an incorrect legal statement.
- Temporal legal reasoning is especially fragile. The overruling-year task shows that models often fail to understand that the legal force of a precedent changes over time.

2.5 Detailed observations

The paper's most important observation is that legal hallucination is not simply a retrieval problem. A model may know that a case exists but still misstate its holding, apply it to the wrong procedural posture, or fail to recognise that it has been overruled. This means that a retrieval system that returns the correct document does not necessarily produce a legally reliable answer.

The paper also shows that legal authority is relational. Questions about whether one case agrees with, distinguishes, limits, or overrules another require reasoning over relationships between authorities, not merely matching text. This is a direct motivation for representing legal authority as a graph rather than as isolated documents.

Finally, the paper demonstrates that legal form matters. A citation may refer to a real case but still be incorrect if the reporter reference, pincite, quotation, or attributed holding is wrong. Therefore, verification must operate at the level of the citation, the quoted span, the proposition, and the temporal status of the authority.

2.6 Limitations

The study focuses on U.S. federal case law and does not evaluate a complete browsing agent operating across live websites. It also focuses on standalone LLM behaviour rather than retrieval-augmented or tool-using systems. Nevertheless, it provides a foundational taxonomy and empirical baseline for legal hallucination research.

3. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools

Authors: Varun Magesh, Faiz Surani, Matthew Dahl, et al.
Publication: Journal of Legal Analysis / arXiv:2405.20362.
Core question: Do retrieval-augmented legal research tools eliminate hallucination, and what types of errors remain?

3.1 Research problem

This paper evaluates whether commercial legal research tools that use retrieval-augmented generation (RAG) actually reduce hallucination. The authors test whether systems that retrieve legal documents before generating an answer are more reliable than general-purpose chatbots.

The paper is important because it distinguishes between a system returning a real source and a system producing an answer that the source actually supports. A legal research tool can retrieve a genuine case or statute while still mischaracterising its holding, applying it to the wrong jurisdiction, or ignoring an exception.

3.2 Data used

The authors construct a preregistered benchmark of more than 200 legal queries. The queries are designed to test different forms of legal research failure rather than merely general legal knowledge.

The benchmark includes four categories of questions:

- **General legal research questions:** broad questions about legal doctrine or legal research.
- **Jurisdiction- and time-specific questions:** questions whose correct answer depends on a particular state, court, or point in time.
- **False-premise questions:** questions that embed an incorrect assumption, testing whether the system identifies and corrects the premise.
- **Factual-recall questions:** questions requiring accurate recall of specific legal facts or authorities.

The systems evaluated are Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law AI, and GPT-4. The commercial tools are included because they represent retrieval-based legal research products rather than general-purpose chatbots.

3.3 Method

Each query is submitted to the evaluated systems, and the resulting response is manually assessed against authoritative legal sources. The authors distinguish between incorrect statements and false claims that a source supports a proposition.

A response is classified as hallucinated if it either states something incorrect or falsely asserts that a source supports a proposition. This definition is broader than simply checking whether a cited document exists. It includes cases in which the document is real but does not support the stated legal conclusion.

The evaluation also examines whether the system identifies an incorrect premise in the user's question. A reliable legal research system should not merely answer a question built on a false assumption; it should flag the assumption and provide the correct legal position.

3.4 Principal findings

- RAG reduces hallucination relative to a general-purpose chatbot, but does not eliminate it. The commercial legal research tools hallucinate between 17% and 33% of the time, while GPT-4 hallucinates 43% of the time in the study.
- Retrieval does not guarantee legal relevance. A system may retrieve a real case or statute that is superficially similar to the query but legally inapplicable because of jurisdiction, date, procedural posture, or a special condition.

- Grounding failures are distinct from factual errors. A response may cite a real authority while misrepresenting what that authority says or supports.
- False premises remain a major weakness. Systems often fail to challenge incorrect assumptions embedded in a user query.
- Commercial claims of hallucination-free performance are overstated. The paper finds that retrieval-based legal tools still produce material rates of hallucination despite their use of authoritative sources.

3.5 Detailed observations

The paper's central contribution is the distinction between source-level and claim-level grounding. A system may successfully retrieve a document, yet fail because the document does not support the particular proposition being asserted. This is especially important in law, where the same case may support one proposition but not another.

The paper also shows that superficial textual similarity is unreliable in legal research. Legal rules often contain conditions, exceptions, definitions, and jurisdictional limits that determine whether they apply. A retrieval system that ranks passages by semantic similarity may return text that looks relevant but is legally inapposite.

The findings support a separation between retrieval, scope checking, temporal checking, and claim verification. Retrieval can identify candidate authorities; scope and temporal checks determine whether the authority applies; and claim verification determines whether the cited text actually supports the proposition.

3.6 Limitations

The study evaluates commercial tools at a particular point in time and does not provide a complete open-source replication of every proprietary system. It also focuses on query answering rather than full browsing-agent trajectories. Nevertheless, it is one of the strongest empirical demonstrations that RAG alone is insufficient for trustworthy legal research.

4. Citation Grounding: Detecting and Reducing LLM Citation Hallucinations via Legal Citation Graphs

Author: Ovcharov.
Publication: arXiv:2606.00898.
Core question: Can a large legal citation graph be used both to detect hallucinated citations and to train models that reduce them?

4.1 Research problem

This paper introduces *Citation Grounding* (CG), a method for verifying whether an LLM-generated legal citation exists in a large citation graph, is contextually relevant, and was valid at the relevant date. It is one of the closest methodological precedents for using a legal graph as a verification oracle and as a source of training signal.

The paper is based on Ukrainian legal materials rather than U.S. law, but its methodological structure is jurisdiction-neutral. It demonstrates that a citation graph can support both evaluation and model training, provided that the graph distinguishes between existence, relevance, and temporal validity.

4.2 Data used

The authors construct a large bipartite legal citation graph from Ukrainian court decisions and statutes. The graph contains:

- 100.8 million decision nodes.
- 21,736 statute nodes.
- 502 million citation edges.

The graph links judicial decisions to the statutes and other decisions they cite. This structure allows the authors to check whether a cited authority exists in the corpus, whether it is contextually relevant to the query, and whether it was valid at the relevant time.

4.3 Method

The paper defines Citation Grounding as a metric with three components: citation precision, citation relevance, and citation temporality. Citation precision asks whether the cited provision exists. Citation relevance asks whether the citation is contextually appropriate. Citation temporality asks whether the provision was valid at the relevant date.

The authors evaluate several systems on 100 Ukrainian legal queries. They compare a RAG system, a fine-tuned model, and a baseline model, measuring how often generated citations survive graph-based verification.

The paper also introduces Citation Grounding DPO (CG-DPO), a training method that constructs preferred and rejected citation pairs by corrupting verified citations. Corruptions include article swaps, law swaps, hallucination injection, and anachronisms. A fine-tuned Qwen2.5-7B model reaches 98.5% validation accuracy in distinguishing correct from corrupted citations.

4.4 Principal findings

- Citation grounding identifies substantial hallucination. Across 100 queries and five systems, CG scores range from 0.791 to 0.873, meaning that 13% to 21% of citations fail graph-based verification.
- RAG improves citation precision. The RAG system achieves the highest CG score while producing fewer citations per response, suggesting that retrieval can improve precision by constraining generation to evidence-backed citations.
- Graph coverage affects evaluation. Some citations classified as hallucinated are real statutes that are absent from the graph because courts rarely cite them directly. This shows that graph coverage must be measured separately from model correctness.

- Temporal validity requires versioned law. The authors note that checking whether a statute exists is weaker than checking whether the exact version was in force on the relevant date.
- Corrupted citations can serve as training data. CG-DPO shows that systematically corrupted citations can generate useful preference pairs for training models to distinguish valid from invalid citations.

4.5 Detailed observations

The paper's most important methodological insight is that a citation graph can serve two roles simultaneously. First, it can act as an evaluation oracle: a generated citation is checked against the graph to determine whether it exists, is relevant, and was valid. Second, it can act as a source of weak supervision: valid citations can be corrupted in controlled ways to create negative examples for training.

The paper also demonstrates that citation verification must distinguish between different types of failure. A citation may fail because it does not exist, because it is irrelevant, or because it was not in force at the relevant time. These are different errors with different remedies.

The finding that RAG produces fewer but more precise citations is important. It suggests that constraining generation to verified sources can reduce hallucination, but it also implies that coverage may decrease if the retrieval corpus is too narrow. This trade-off between precision and coverage is central to any verified-source registry design.

4.6 Limitations

The study is based on Ukrainian law and uses a citation graph rather than a full legal-agent system. It does not evaluate browsing, authentication, scope frames, or contract drafting. Nevertheless, its graph-based verification and corruption-based training methods are highly transferable to other legal domains.

5. PARCEL: When Citations Mislead? A Claim-Level Benchmark for Legal Hallucination Detection

Authors: M. Mikail Demir and M. Abdullah Canbaz.
Publication: ICAIL 2026; arXiv:2610.10971.
Core question: Can LLMs reliably determine whether a legal claim is supported by the authority cited for it?

5.1 Research problem

PARCEL focuses on a narrower and more difficult problem than document retrieval: whether a specific legal claim is supported by the specific authority cited for it. The paper argues that legal hallucination often takes the form of a confident claim paired with a citation that does not actually support the proposition.

The paper uses the explanatory parenthetical as its unit of analysis. An explanatory parenthetical is a short explanation attached to a legal citation, such as “holding that…” or “finding that…”. These parentheticals are used by judges, clerks, and attorneys to convey the relevance of an authority to a legal proposition.

5.2 Data used

The dataset is constructed from recent New York State Court of Appeals decisions issued between 2024 and 2025. The authors retrieve 85 publicly available briefs filed before the New York State Court of Appeals through the CourtPASS system.

From these briefs, the authors extract explanatory parenthetical citations and their associated cases. They obtain more than 700 candidate examples. After filtering out direct quotations and surface-level factual summaries, they retain 180 high-quality claims. These claims are used as few-shot exemplars for synthetic claim generation.

The authors then generate synthetic claims directly from 85 judicial decisions. Each decision is used to generate up to ten supported claims, ensuring that each usage type appears at least twice. This produces 849 supported claims. From these, the authors generate refuted and not-found claims, producing a final dataset of 3,396 claims with a controlled 1:2:1 class balance across Supported, Refuted, and Not Found.

5.3 Method

The paper casts parenthetical verification as a three-way natural language inference (NLI) task. Given a judicial decision and a parenthetical claim, the model must classify the claim as Supported, Refuted, or Not Found.

The authors use a usage-type rubric to generate supported claims. The rubric includes five types: holding rule, reasoning rationale, facts procedure, treatment relational, and descriptive label. This ensures that the synthetic claims cover different ways in which legal authorities are actually used in legal writing.

Refuted claims are generated using four strategies: distractor entity swap, logic reversal, scope contradiction, and authority swap. Not-found claims are generated using three strategies: phantom citation, unaddressed argument, and missing detail. All adversarial claims are generated while conditioning on the full text of the judicial decision, so that perturbations remain stylistically consistent with the source.

The evaluation is zero-shot: each model is given the full judicial decision as the premise and a single parenthetical claim as the hypothesis. The model must rely on its internal representations, long-context processing, and implicit legal reasoning to perform the verification.

5.4 Principal findings

- Modern LLMs can perform claim-level legal verification with high apparent accuracy. The strongest model, Gemini 3 Flash Preview, reaches 97% overall accuracy and 0.97 macro F1.
- Models still incorrectly mark unsupported claims as supported. Even when the full opinion text is provided, models sometimes validate claims that the opinion does not support.
- Missing support is harder to detect than direct contradiction. The Not Found class is the main bottleneck: models are better at recognising explicit contradictions than at recognising when supporting evidence is

absent.

- Fabricated but plausible citations cause the largest performance drop. Phantom citations are the most difficult adversarial strategy, with average accuracy of 0.72 across models.
- Parametric legal knowledge can override textual grounding. Models sometimes validate claims that are legally accurate in the abstract but unsupported by the specific judicial decision provided.

5.5 Detailed observations

The paper introduces the concept of a False Entailment Rate (FER), defined as the probability that a model predicts entailment when the ground truth is either refuted or not found. This metric captures a safety-relevant failure: the model asserts legal support where none exists.

The paper finds a consistent asymmetry: false entailment on not-found claims is substantially higher than false entailment on refuted claims. For example, GLM-4.7 achieves a false entailment rate of 1.18% on refuted claims but 9.55% on not-found claims. This means that models are better at detecting explicit contradictions than at detecting absence of support.

The paper also identifies an “inverse contamination effect.” Rather than benefiting from memorised legal knowledge, models’ pre-trained legal knowledge can interfere with closed-book verification by overriding textual constraints. A claim may be legally correct in general but still constitute a hallucination if it is not supported by the cited authority.

The paper’s error analysis shows that most hard claims are not universally difficult. Only 8 claims out of 3,396 are misclassified by all seven models. This suggests that ensemble-based verification or verifier cascades could reduce hallucination risk, since different models have different blind spots.

5.6 Limitations

The study is limited to New York State Court of Appeals decisions and uses synthetic claim generation. It does not evaluate a full browsing agent or a complete legal research pipeline. Nevertheless, it provides a rigorous framework for claim-level grounding evaluation that can be adapted to other jurisdictions and document types.

6. LexAgentHallu: A Hierarchical Benchmark for Profiling Hallucinations in Legal Agents

Authors: Yujin Zhou, Mingxuan Zheng, Chuxue Cao, Yidan Huang, Jiale Chen, Yike Guo, and Sirui Han.
Publication: EMNLP 2026; arXiv:2609.09754.
Core question: How do legal agents hallucinate across multi-step trajectories, and can errors be attributed to specific steps?

6.1 Research problem

LexAgentHallu is the first benchmark dedicated to evaluating hallucinations in legal LLM agents operating across multi-step trajectories. It argues that existing legal benchmarks evaluate single-turn QA with outcome-level metrics, while agentic hallucination benchmarks lack legal-specific diagnostic capability.

The paper identifies a key gap: when a legal agent produces a hallucinated conclusion, it is often unclear whether the error stems from flawed legal knowledge or from a faulty agentic process. LexAgentHallu addresses this by evaluating both substantive legal errors and agent-procedural errors.

6.2 Data used

The benchmark contains 3,414 expert-curated instances across 17 legal categories and 6 task types. The data is drawn from two complementary streams: open-source legal benchmarks and real-world practice materials from China's National Judicial Examination (Fakao).

The open-source stream draws from LexEval, LawBench, UniLaw, DISC-Law-Eval, and PLawBench. The real-world-practice stream consists of items manually collected from recent National Judicial Examination releases.

The benchmark uses a four-stage expert-in-the-loop pipeline: multi-source data collection, data filtering, expert annotation and verification, and taxonomy-driven rubric checklist generation. Each instance is paired with a taxonomy-anchored rubric checklist that enables fine-grained evaluation of hallucination subclasses.

6.3 Method

The paper uses a dual-layer hallucination taxonomy with 7 mid-level categories and 27 fine-grained subclasses. Layer 1 captures substantive legal errors, including authority hallucination, doctrine hallucination, procedural-law hallucination, and application/subsumption hallucination. Layer 2 captures agent-procedural errors, including planning/reasoning, memory, and tool-call/observation errors.

The evaluation uses a rubric-anchored LLM-as-a-judge procedure with seven dedicated prompts, one per top-level taxonomy category. Each judge call focuses on a narrow, coherent error family rather than the full 27-subclass space. The judge operates on the full trajectory rather than the final answer alone.

The paper introduces several metrics: hallucination frequency (HF), hallucination density (HD), answer correctness, substantive/procedural cleanliness, and Right-Answer-Wrong-Reason (RAWR) rate. These metrics quantify both how often and how broadly each rollout hallucinates.

6.4 Principal findings

- Hallucination is pervasive across all systems. Even the best-performing configuration, Gemini-3.1-Pro under ReAct, triggers at least one hallucination in 89.0% of rollouts.
- Framework-dependent hallucination profiles. Model rankings shift across frameworks, indicating that no single orchestration strategy universally dominates.
- Right-Answer-Wrong-Reason effect. Even when a legal agent produces a correct final answer, its reasoning trajectory still contains substantive-legal hallucinations (68% on average) and agent-procedural hallucinations (37% on average).
- Hallucination subclasses cluster. Errors are not independent slips but structured failure modes rooted in shared mechanisms.

- Scaling behaviour. Performance correlates positively with model scale, but frontier closed-source models still outperform the largest open-source variants on most metrics.

6.5 Detailed observations

The paper's most important finding is the Right-Answer-Wrong-Reason effect. A legal agent may reach a superficially correct conclusion while relying on fabricated, misread, or irrelevant intermediate evidence. This means that final-answer accuracy is an unreliable proxy for legal faithfulness.

The paper also shows that hallucination subclasses cluster rather than scatter. Specific mechanistic pairs rise sharply above chance co-occurrence. For example, temporal/limitation errors co-occur with remedy-path errors, and procedural-step errors co-occur with procedural-consequence errors. This suggests that detecting high-frequency hallucination types can serve as early signals to flag correlated low-frequency counterparts.

The paper's taxonomy is particularly useful because it separates substantive legal errors from agent-procedural errors. This distinction allows researchers to identify whether a failure stems from legal knowledge, planning, memory, or tool use. It also enables targeted interventions at specific points in the agent pipeline.

6.6 Limitations

The benchmark is grounded in the Chinese legal system, with all queries, gold trajectories, and rubric checklists written in Chinese and anchored to PRC statutes, judicial interpretations, and procedural rules. This limits cross-jurisdictional generalizability. The authors acknowledge this limitation and plan to extend the taxonomy to other legal systems.

7. Who Checks the Citations? Benchmarking Legal Hallucination Detection

Authors: Patty Liu, Dominik Stammbach, and Peter Henderson.
Publication: arXiv:2606.21155.
Core question: Can AI systems automatically detect hallucinated legal citations in court filings?

7.1 Research problem

This paper examines whether AI systems can detect hallucinated legal citations in court filings. It is motivated by the growing number of court filings containing fabricated citations, and by the observation that hallucination rates do not consistently decrease across model generations.

The paper identifies over 1,000 court filings containing hallucinated citations, with this number growing year-over-year. It argues that the problem is not self-correcting and that automated verification tools are needed to reduce the burden on courts, attorneys, and litigants.

7.2 Data used

The paper introduces LePhantomCite, a benchmarking dataset of 1,300 legal brief excerpts with injected hallucinations. The dataset is drawn from two sources:

- 1,000 excerpts from real appellate briefs with systematically injected hallucinations, covering five hallucination types.
- 300 entries from the LLM-generated central holdings portion of Dahl et al. (2024), manually verified and reformatted.

The appellate brief excerpts are collected from 245 federal appellate briefs filed in 13 U.S. Courts of Appeals between January 2012 and December 2021, retrieved via the CourtListener API. The briefs are restricted to pre-2022 filings to minimise the risk that the source documents themselves contain AI-generated content.

The briefs are converted from PDF to plain text using olmOCR, parsed into sentences using a fine-tuned RoBERTa sentence segmentation model, and grouped into semantically coherent segments using Llama-3.3-70B-Instruct. This yields 5,648 segments, from which 1,000 are subsampled.

For each citation in the sampled segments, the authors use Qwen3-32B to extract the associated case name, quotation, and holding proposition, constrained to exact substrings of the original text. Hallucinations are then injected by systematically modifying citation components according to a taxonomy derived from real court filings.

7.3 Method

The paper defines five categories of legal citation hallucination: non-existent citation, mismatched case name, incorrect pincite, verbatim misquote, and content misrepresentation. Each category represents a different type of failure in the citation-verification process.

The authors evaluate five models with an agentic verification system built on the Bayesian Optimal Experimental Design (BOED) agent framework. The agent maintains an explicit, language-based belief state that is updated after each action. This belief state tracks all case citations, quotations, and holdings extracted from the input excerpt, along with the agent's current assessment of each element.

The agent has access to eight actions, including legal database search and web search. The evaluation uses a 70-30 train-test split, with all evaluations run on the test set of 390 examples. The paper also evaluates a non-agentic baseline and a production agent harness (Claude Code with Opus 4.8).

7.4 Principal findings

- Agentic verification improves recall. The BOED agentic harness achieves higher recall than non-agentic experiments across all models.
- GPT-5 achieves the highest recall. GPT-5 achieves 84.4% recall and 55.0% F1 in the agentic framework, outperforming all other models.

- All models struggle with subtle error categories. No model can reliably detect incorrect pincites, and content misrepresentation remains difficult.
- Information access is a binding constraint. Restricted access to legal databases limits the efficacy of even the best agents.
- Agent verification is resource-intensive. GPT-5 averages 15.3 steps per excerpt, and the agent sometimes reaches the maximum step limit before verifying all citations.

7.5 Detailed observations

The paper's error analysis reveals several important failure modes. First, weaker models tend to treat absence from CourtListener as evidence of hallucination, producing false positives. Stronger models attempt to gather more information from alternative sources before reaching a verdict.

Second, agents often exhaust their verification budget due to redundant actions. GPT-5 hits the step limit on 29.5% of test episodes, and duplicate citation lookups or opinion searches appear in a substantial fraction of episodes. This suggests that agent efficiency is as important as agent capability.

Third, the paper finds that page-number information for many cases is only accessible through commercial databases such as Westlaw and LexisNexis. This explains why models perform worst on incorrect pincites, and it highlights a structural limitation of publicly available legal data.

7.6 Limitations

The study focuses on U.S. federal appellate briefs and uses injected hallucinations rather than naturally occurring errors. It also does not evaluate the full range of legal research tasks, such as contract drafting or statutory interpretation. Nevertheless, it provides a rigorous framework for citation verification that can be adapted to other legal domains.

8. Cross-paper synthesis

The six papers collectively establish that legal hallucination is a multi-layered problem that cannot be solved by retrieval alone. The following table summarises the main findings and their implications for graph-based verification.

8.1 Summary of findings

Paper	Core finding	Implication for graph/GNN
Large Legal Fictions	Legal hallucinations are pervasive; models fail on temporal and doctrinal reasoning	Graph must encode temporal validity and doctrinal relationships.
Hallucination-Free?	RAG reduces but does not eliminate hallucination; grounding failures persist	Separate retrieval, scope, temporal, and claim verification.
Citation Grounding	Citation graphs can verify existence, relevance, and temporality; graph-based retrieval can provide training signal	Use graph structure as both evaluation and training source of weak supervi
PARCEL	Missing support is harder to detect than contradiction; fabricated authorities are hardest	Target absence detection, not just contradiction.
LexAgentHallu	Agent errors must be localised across multi-step trajectories; evaluate at trajectory level, not document level.	Evaluate trajectories, not just final answers; use step-level attribu
Who Checks the Citations?	Agentic verification improves recall but is resource-intensive; access is a first-order concern	Use GNN as triage model; reserve expensive verification for unc

8.2 Common themes

Across all six papers, several themes emerge. First, legal hallucination is not a single failure mode but a family of related errors that require different detection strategies. Second, retrieval improves performance but does not guarantee correctness. Third, legal authority is relational, and graph-based methods are well suited to capturing these relationships. Fourth, evaluation must operate at multiple levels: document, claim, citation, and trajectory.

The papers also converge on the importance of provenance and access. A system that cannot access authoritative sources is more likely to substitute unverified alternatives, and a system that cannot record provenance cannot be audited. This supports the design of a verified-source registry with content hashing, session tracking, and provenance logging.

8.3 Implications for GNN training

The literature supports a specific architecture for GNN training in legal research. The GNN should not be trained to “know the law” in the ordinary sense. Instead, it should learn relational regularities in a versioned legal graph, acting as an advisory detector of implausible or incomplete legal claims.

The recommended architecture is:

- **Input:** typed nodes (statutes, provisions, cases, holdings, courts, transaction types, actor classes, process facts, source records, claims) and typed edges (cites, interprets, follows, distinguishes, limits, overrules, applies_to, valid_from, valid_to, retrieved_by, supports).
- **Training tasks:** authority-existence validation, temporal link prediction, scope-aware authority matching, claim-support plausibility, missing-authority gap ranking, and access-risk prediction.
- **Output:** advisory alert and ranked list of potentially missing authorities; never overrides deterministic guards.
- **Evaluation:** compare against text-only, homogeneous GNN, and citation-grounding baselines; report per-failure-mode metrics rather than aggregate accuracy.

8.4 Concrete GNN training plan

Based on the literature, the following training plan is recommended:

- **Task 1: Authority-existence and registry validation.** Detect fabricated or unregistered authorities using positive examples from the verified registry and negative examples from nonexistent or malformed citations.
- **Task 2: Temporal link prediction.** Predict whether an authority was valid, controlling, or applicable at a given date, using temporal splits and amendment/repeal/overruling edges.

- **Task 3: Scope-aware authority matching.** Predict whether an authority applies to a matter's actor and transaction type, using positive and negative examples based on scope frames.
- **Task 4: Claim-support plausibility.** Predict whether a cited authority plausibly supports a legal proposition, using PARCEL-style labels and combining GNN structural features with text embeddings.
- **Task 5: Missing-authority gap ranking.** Suggest authorities the agent may have missed, trained on citation and treatment edges.
- **Task 6: Access-risk prediction.** Predict whether a source is likely to be blocked, paywalled, stale, or unofficial, using features such as domain, URL pattern, and retrieval history.

8.5 Recommended graph schema

The following node and edge types are recommended for the legal knowledge graph:

- **Nodes:** Statute, ProvisionVersion, Case, Holding, Court, TransactionType, ActorClass, ProcessFact, SourceRecord, Claim.
- **Edges:** cites, interprets, follows, distinguishes, limits, overrules, superseded_by_statute, applies_to, valid_from, valid_to, retrieved_by, supports.

This schema supports both deterministic guards (source, scope, temporal) and learned GNN signals (plausibility, gap ranking, access risk). It also enables point-in-time queries, which are essential for temporal reasoning in law.

8.6 Final synthesis

The literature does not support training one GNN on a giant legal corpus and treating its output as ground truth. Instead, it supports a narrower, more defensible design: a live browsing agent whose outputs cannot enter the dossier unless they survive independently checkable source, scope, temporal, and textual-support gates; the GNN improves recall and flags suspicious relational patterns, but never becomes the source of legal truth.

The strongest version of the contribution is therefore: a live browsing agent operating under source, scope, temporal, claim-support, provenance, and access constraints, with a heterogeneous GNN used only as an advisory plausibility and gap-ranking signal. This design is supported by all six papers reviewed in this report.